Take-Home Assignment

Conformal prediction for regression

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Objective

The objective of this assignment is to study the use of conformal prediction (CP) for uncertainty quantification in a regression problem. The work can be completed in groups of 2 to 5 students. You will select a real-world regression dataset, compare two machine learning (ML) regression models, and investigate how different CP methods affect the resulting prediction intervals.

Data

Choose a real-world regression dataset. The **UCI Machine Learning Repository** is suggested as a possible source of datasets, but you are free to use data from **any other relevant and reliable source**. The dataset should contain a continuous target variable and a sufficient number of observations for training, calibration and testing. You will briefly describe the context and objective of the dataset, the target variable and main predictors, the number of observations and variables, and the main preprocessing steps, if any. **All data sources must be clearly cited and referenced in the report.**

Machine learning models

Choose two different regression algorithms. At least one relatively simple model, such as linear regression or a simple deicsion tree, is encouraged. The second model can be a more flexible (and less interpretable) ML method, such as random forest, or a neural network. Briefly explain the main characteristics of the two models and compare their predictive performance using appropriate metrics such as RMSE, MAE or R^2.

Conformal prediction

Apply two different CP methods, among split CP, Jackknife, Jackknife+, CV or another appropriate CP method. For each method, briefly explain the principle, the calibration procedure and the construction of the prediction intervals. The analysis should compare the four combinations (2 ML models $\times$ 2 CP methods). Evaluate the resulting prediction intervals in terms of regression coverage score and mean width score. You may also consider other relevant measures or graphical representations.

Results and Discussion

The report should focus on the interpretation of the empirical results. You will discuss in particular the differences between the ML models and CP methods, whether the observed coverage is consistent with the nominal coverage (do you reach the $1 - \alpha$ guarantee?), the width of the prediction intervals, and the potential relationship between predictive accuracy and uncertainty quantification.

Final report

The final deliverable is a PDF report of approximately 10–12 pages, excluding references and optional appendices. The report should be written as a short empirical research paper and follow the following structure:

1. Abstract
2. Introduction
3. Methods & Data
4. Results
5. Discussion
6. Conclusion
7. References
8. Appendix (optional)

The report should be self-contained and clearly present the methodology, empirical results and interpretation. **No code is required in the report.**